Five-row arrays increasing in four directions: a proof of the Kauers–Koutschan formula for OEIS A181199

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Abstract

Let a_n be the number of fillings of a $5 \times n$ rectangle with $1, \dots, 5n$ that increase along rows, columns, diagonals and downward antidiagonals (OEIS A181199). Kauers and Koutschan conjectured an explicit nested hypergeometric sum for a_n (their Conjecture 19), equivalent to an order-three recurrence of degree 24 recorded in the OEIS. We prove both statements for every $n \geq 1$.

The proof starts from the order polytope of the array. Its columns form a chain of interlacing 5-tuples. Cauchy–Binet and de Bruijn's Pfaffian formula turn the volume into a 6×6 Pfaffian with an explicit polynomial kernel. Integrating over the last column gives a double sum of hypergeometric terms with one rational kernel. An explicit telescoping certificate shows that $T(n+1,k,l) - T(n,k,l)$ is a two-dimensional difference, which reduces $a_{n+1} - a_n$ to a single sum. A Gosper certificate then proves the first-order relation that underlies the nested sum.

All identities are exact. The two large certificates are supplied as data and checked by exact polynomial arithmetic; direct enumeration of the arrays serves only as an independent consistency check.

1 The problem and the result

For $n \geq 1$, an *admissible filling* is a bijection $u\colon \{1,\dots,5\} \times \{1,\dots,n\} \to \{1,\dots,5n\}$ such that

$$u(i,j) < u(i',j') \quad \text{whenever} \quad (i'-i, j'-j) \in \{(0,1),(1,0),(1,1),(1,-1)\}$$

and both cells exist. Let a_n be the number of admissible fillings. The sequence is OEIS A181199 [4] (offset 1); it begins

$$a_1, \dots, a_7 = 1,\ 1,\ 16,\ 985,\ 141696,\ 36372976,\ 14083834704.$$

The diagonal condition $(1,1)$ follows from the others, but we keep it in the definition.

Throughout, $(x)^{\overline{m}} = x(x+1)\cdots(x+m-1)$ denotes the rising factorial. Put

$$\begin{aligned} q(t) &= 25216t^8 + 9888t^7 - 14496t^6 + 11208t^5 + 23832t^4 + 7383t^3 - 1522t^2 - 939t - 90, \\ p(t) &= 137855872t^{11} + 860969696t^{10} + 2047036856t^9 + 2032587274t^8 - 24192441t^7 \\ &\quad - 1894061166t^6 - 1671661480t^5 - 524330624t^4 + 36004789t^3 + 62751860t^2 \\ &\quad + 13865604t + 927360, \end{aligned}$$

and

$$u(k) = \frac{8\,q(k)}{(2k-1)(4k-1)(3k+1)^{\overline{3}}(4k+1)^{\overline{4}}}, \tag{1.1}$$

$$v(i) = \frac{(3i+1)(3i+2)(4i+3)\,p(i)}{(i+1)^2(i+2)^2(2i-1)(2i+1)(2i+3)\,q(i)\,q(i+1)}.$$

The second octic in the original statement of [3] is $q(i+1)$ expanded. For integers $k, i \geq 1$, all linear factors in (1.1) are positive. Moreover, for real $t \geq 1$, $q(t) \geq 25216t^8 - 14496t^6 - 2551t^2 \geq 8169t^2 > 0$. Likewise $p(t) > 0$: the coefficients of $t^{11}, \dots, t^8$ sum to 5078449698, the negative coefficients sum to -4114245711, and the lower positive terms are positive. Hence $u(k)$ and $v(i)$ are well defined and nonzero for all $k, i \geq 1$; no singularity occurs in the range of summation. Define the hypergeometric terms

$$U(k) = -\frac{27}{4}(-1)^k u(k) \frac{(5k)!}{(3k)!\,k!^2}, \qquad V(i) = (-1)^i v(i) \frac{(3i)!}{i!^3}. \tag{1.2}$$

Theorem 1 (Kauers–Koutschan Conjecture 19)**.** *For every $n \geq 1$,*

$$a_n = 1 - \frac{27}{4} \sum_{k=1}^{n-1} (-1)^k u(k) \frac{(5k)!}{(3k)!\,k!^2} \sum_{i=1}^{k-1} (-1)^i v(i) \frac{(3i)!}{i!^3} = 1 + \sum_{k=1}^{n-1} U(k) \sum_{i=1}^{k-1} V(i). \tag{1.3}$$

Corollary 2 (the OEIS recurrence)**.** *For every $n \geq 1$, $\sum_{j=0}^{3} p_j(n)\, a_{n+j} = 0$, where $p_0, \dots, p_3$ are the degree-24 polynomials of Appendix B, the recurrence contributed to A181199 by C. Koutschan. Its leading coefficient has no zero at an integer $n \geq 1$. Hence, together with $a_1 = a_2 = 1$ and $a_3 = 16$, the recurrence determines the sequence.*

Context. The sequence was introduced by R. H. Hardin in 2010. Kauers and Koutschan [3, Section 6.4] computed 60 terms from a profile walk (Lemma 3 below) and guessed the recurrence by LLL-based guessing. They observed that Maple factors its operator into three first-order operators, which suggested (1.3); Table 1 of [3] marks the case as open. A 2026 survey talk by Kauers [2] still lists it with a question mark. Sun [7] identified these arrays with standard Young tableaux of truncated shifted strips. Using integrals over order polytopes, he proved recurrences for three rows and for the other parameter direction (fixed row length 4 or 5). Our Section 2 follows his volume viewpoint but replaces his inductive evaluation by Cauchy–Binet and de Bruijn's formula. The four-row sequence A181198 is a different problem and is not used here.

Scope. Theorem 1 and Corollary 2 are proved for all $n \geq 1$ with the OEIS offset; the value $n = 0$ is never used. A bounded literature search (31 October 2026) found no earlier proof. This is evidence, not a priority claim.

Plan. Section 2 writes $a_n/(5n)!$ as an integral of a 6×6 Pfaffian. Section 3 turns it into an explicit double sum (Theorem 9). Section 4 gives the two-dimensional telescoping relation, and Section 5 the single-sum identity. Section 6 assembles the proof, and Section 7 describes the computations and checks.

2 Profiles and the order polytope

Lemma 3 (profiles)**.** *Write x_i for the number of cells of row i that carry the m smallest labels. The admissible fillings correspond bijectively to lattice walks from $(0, \dots, 0)$ to $(n, \dots, n)$ in $\mathbb{Z}^5$ with unit steps. Every visited point satisfies $0 \leq x_i \leq n$ and $x_{i-1} \geq \min(n, x_i + 1)$ whenever $x_i > 0$.*

Proof. The set of cells with the m smallest labels is an order ideal. Row increase makes each row a prefix. The cell (i, x_i) requires $(i-1, x_i)$ and, if $x_i < n$, also $(i-1, x_i + 1)$. Conversely these inequalities imply all predecessor conditions. □

This lemma is used only for the independent computer checks of Section 7.

Let $\mathcal{O}_n \subset [0,1]^{5\times n}$ be the order polytope: points y with $y_c < y_{c'}$ whenever cell c must carry a smaller entry than c'.

Lemma 4 (Stanley [6])**.** $a_n = (5n)!\,\mathrm{vol}(\mathcal{O}_n)$.

Proof. Let $Y \in [0,1]^{5\times n}$ have independent uniform entries. Almost surely the entries are distinct and induce a uniformly random bijection to $\{1,\dots,5n\}$ by ranking. $Y \in \mathcal{O}_n$ exactly when that ranking is an admissible filling, so $\mathrm{vol}(\mathcal{O}_n) = a_n/(5n)!$. □

For $x, y \in \mathbb{R}^5$ write $x \prec y$ if $x_1 < y_1 < x_2 < y_2 < \cdots < x_5 < y_5$.

Lemma 5 (interlacing columns)**.** *Let $n \geq 2$ and let $c^{(j)} = (y_{1j},\dots,y_{5j})$ denote the j-th column. Up to a null set,*

$$\mathcal{O}_n = \{y \in [0,1]^{5\times n} :\ c^{(1)} \prec c^{(2)} \prec \cdots \prec c^{(n)},\ y_{11} > 0,\ y_{5n} < 1\}.$$

Proof. For consecutive columns, the relations of the poset are $y_{i,j} < y_{i,j+1}$ (rightwards) and $y_{i,j+1} < y_{i+1,j}$ (down-left from $(i,j+1)$). Together they say $c^{(j)} \prec c^{(j+1)}$. The remaining relations follow from these. Vertical order: $y_{i,j} < y_{i,j+1} < y_{i+1,j}$ for $j < n$, and $y_{i,n} < y_{i+1,n-1} < y_{i+1,n}$. Down-right: $y_{i,j} < y_{i,j+1} < y_{i+1,j+1}$. All coordinates lie between y_{11} and y_{5n}. □

For $N \geq 1$ and $0 < t_1 < \cdots < t_5 < 1$ let

$$J_N(t) = \mathrm{vol}\{(c^{(1)},\dots,c^{(N)}) \in (\mathbb{R}^5)^N :\ c^{(1)} \prec \cdots \prec c^{(N)} \prec t,\ c_1^{(1)} > 0\},$$

the volume of the first N columns when the last column equals t. By Lemmas 4 and 5, for $n \geq 2$,

$$a_n = (5n)! \int_{0<t_1<\cdots<t_5<1} J_{n-1}(t)\,\mathrm{d}t. \tag{2.1}$$

Lemma 6 (interlacing determinant)**.** *For $x_1 < \cdots < x_5$ and $y_1 < \cdots < y_5$,* $\det[\mathbf{1}(x_i < y_j)]_{i,j=1}^5 = \mathbf{1}(x \prec y)$.

Proof. Column j of the matrix is $(1,\dots,1,0,\dots,0)^{\mathsf{T}}$ with $r_j = \#\{i : x_i < y_j\}$ ones, and $r_1 \leq \cdots \leq r_5$. The determinant is nonzero only if the columns are distinct and nonzero, i.e. $1 \leq r_1 < \cdots < r_5 \leq 5$, hence $r_j = j$. The matrix is then upper unitriangular, with determinant 1. Finally $r_j = j$ for all j is exactly $x \prec y$. □

Lemma 7. *For $N \geq 1$ put $\varphi_N(t,x) = (t-x)_+^{N-1}/(N-1)!$, where $\varphi_1(t,x) = \mathbf{1}(x < t)$. Then*

$$J_N(t) = \int_{0<x_1<\cdots<x_5} \det\big[\varphi_N(t_j,x_i)\big]_{i,j=1}^5\,\mathrm{d}x.$$

Proof. By Lemma 6, $J_N(t)$ is the integral of $\prod_{m=1}^N \det[\mathbf{1}(c_i^{(m)} < c_j^{(m+1)})]$, where $c^{(N+1)} = t$. The integral runs over ordered $c^{(1)}$ with $c_1^{(1)} > 0$ and over ordered $c^{(2)},\dots,c^{(N)} \in \mathbb{R}^5$; positivity and the bound $t_5 < 1$ of the later columns are automatic. The continuous Cauchy–Binet (Andréief) identity

$$\int_{y_1<\cdots<y_5} \det[K(x_i,y_j)]\det[L(y_i,z_j)]\,\mathrm{d}y = \det\Big[\int_{\mathbb{R}} K(x_i,y)L(y,z_j)\,\mathrm{d}y\Big]$$

eliminates $c^{(2)},\dots,c^{(N)}$ one at a time. The kernel becomes $\varphi_1 * \cdots * \varphi_1$ (N factors), and $\int \varphi_m(y,x)\mathbf{1}(y<t)\,\mathrm{d}y = \varphi_{m+1}(t,x)$. □

Lemma 8 (Pfaffian form)**.** *For $N \ge 1$ and $0 < a < b$ put $\beta_N(a) = a^N/N!$ and*

$$\Psi_N(a,b) = \sum_{k=0}^{N} w_k \, \frac{(-1)^k a^{N+k} b^{N-k}}{(N-k)!\,(N+k)!}, \qquad w_0 = 1,\ w_k = 2\ (k \ge 1). \tag{2.2}$$

Let $M(t)$ be the antisymmetric 6×6 matrix with $M_{pq} = \Psi_N(t_p, t_q)$ for $1 \le p < q \le 5$ and $M_{r6} = \beta_N(t_r)$. Then $J_N(t) = \operatorname{Pf} M(t)$ for $0 < t_1 < \cdots < t_5$.

Proof. De Bruijn's formula [1] for an odd number m of functions reads

$$\int_{x_1 < \cdots < x_m} \det[\phi_j(x_i)]\,\mathrm{d}x = \operatorname{Pf} \begin{pmatrix} A & \beta \\ -\beta^{\mathsf T} & 0 \end{pmatrix},$$

$$A_{pq} = \iint \operatorname{sgn}(y-x)\phi_p(x)\phi_q(y)\,\mathrm{d}x\,\mathrm{d}y, \qquad \beta_p = \int \phi_p.$$

We apply it with $\phi_j(x) = \varphi_N(t_j, x)\mathbf{1}(x > 0)$; then $\beta_j = t_j^N/N!$. For $a < b$ write $F_b(x) = \int_x^\infty \phi_b = (b-x)_+^N/N!$. Splitting the region $x < y$ gives

$$A_{ab} = \int_0^a \phi_a(x) F_b(x)\,\mathrm{d}x - \int_0^a \phi_a(y)\big(F_b(0) - F_b(y)\big)\,\mathrm{d}y = 2\int_0^a \phi_a F_b\,\mathrm{d}x - \beta_N(a)\beta_N(b).$$

Substituting $x = as$ and expanding $(b - as)^N$ yields

$$\int_0^a \frac{(a-x)^{N-1}(b-x)^N}{(N-1)!\,N!}\,\mathrm{d}x = \sum_{k=0}^{N} \frac{(-1)^k a^{N+k} b^{N-k}}{(N-k)!(N+k)!},$$

using $\int_0^1 (1-s)^{N-1} s^k\,\mathrm{d}s = (N-1)!k!/(N+k)!$. The $k = 0$ term of twice this sum, minus $a^N b^N/N!^2$, gives $w_0 = 1$. Hence $A_{ab} = \Psi_N(a,b)$. □

The program `pfaffian_sum.py` independently confirms Lemma 8 for $1 \le N \le 6$. It computes J_N by iterated polynomial integration $J_N(t) = \int_{u \prec t} J_{N-1}(u)\,\mathrm{d}u$ (Sun's recursion) and compares the result with the Pfaffian.

3 An explicit double sum

Let $N = n - 1$. There are 15 perfect matchings of $\{1, \dots, 6\}$. Write a matching as

$$\pi = \{(r,6), (p_1,q_1), (p_2,q_2)\}, \qquad p_i < q_i,$$

and let $\operatorname{sgn}\pi$ be its Pfaffian sign. Expanding the Pfaffian, each term of $J_N(t)$ is a monomial t^e with

$$e_r = N, \quad e_{p_1} = N + k, \quad e_{q_1} = N - k, \quad e_{p_2} = N + l, \quad e_{q_2} = N - l,$$

and coefficient $w_k w_l \chi(k)\chi(l)/N!$, where $\chi(k) = (-1)^k/((N-k)!(N+k)!)$. Moreover

$$\int_{0 < t_1 < \cdots < t_5 < 1} t^e\,\mathrm{d}t = \prod_{s=1}^{5} E_s^{-1}, \qquad E_s = \sum_{i \le s} (e_i + 1).$$

Summing the fifteen resulting rational functions (listed in Appendix A) and symmetrising in $k \leftrightarrow l$ gives, in terms of $n = N + 1$,

$$\mathcal{R}(n,k,l) = \frac{P(n,k,l)}{120\,n^3 D_n(k) D_n(l)(2n+k+l)}, \qquad D_n(x) = (x+n)(x+2n)(x+3n)(x+4n). \tag{3.1}$$

Here P is the symmetric form of degree 7

$$\begin{aligned}P(n,k,l) &= 1152n^7 + 2976n^6(k+l) + 2400n^5(k^2+l^2) + 7400n^5kl + 840n^4(k^3+l^3)\\ &\quad + 5600n^4kl(k+l) + 120n^3(k^4+l^4) + 1870n^3kl(k^2+l^2) + 3598n^3k^2l^2\\ &\quad + 250n^2kl(k^3+l^3) + 999n^2k^2l^2(k+l) + 115nk^2l^2(k^2+l^2)\\ &\quad + 200nk^3l^3 + 15k^3l^3(k+l).\end{aligned}$$

Define, for $n \geq 1$ and integers k, l,

$$\chi_n(k) = \frac{(-1)^k}{(n-1-k)!\,(n-1+k)!}, \qquad g(n) = \frac{(5n)!}{120\,n^3(n-1)!},$$
$$T(n,k,l) = g(n)\chi_n(k)\chi_n(l)\frac{P(n,k,l)}{D_n(k)D_n(l)(2n+k+l)}, \tag{3.2}$$

with $1/m! = 0$ for negative integers m. Thus $T(n,k,l) = 0$ unless $0 \leq k, l \leq n-1$ (for $k, l \geq 0$).

Theorem 9 (double sum)**.** *For every $n \geq 1$, $a_n = \sum_{k=0}^{n-1}\sum_{l=0}^{n-1} w_k w_l\, T(n,k,l)$.*

Proof. For $n \geq 2$ combine (2.1), Lemma 7, Lemma 8, the monomial integration above and the symmetry of the weights $w_k w_l$. The prefactor is $(5n)!/N! = 120\,n^3 g(n)$. For $n = 1$ the right side is $T(1,0,0) = 1 \cdot 1152/(24^2 \cdot 2) = 1 = a_1$. □

All later steps are identities between hypergeometric terms with this kernel. Put

$$D(n) = \sum_{k,l\geq 0} T(n,k,l), \qquad B(n) = \sum_{l\geq 0} T(n,0,l).$$

Since T is symmetric in (k,l) and $w_k = 2 - [k=0]$,

$$a_n = 4D(n) - 4B(n) + T(n,0,0) \qquad (n \geq 1). \tag{3.3}$$

4 A two-dimensional telescoping relation

Let $H(m,k,l) = g(m)\chi_m(k)\chi_m(l)$; it vanishes unless $0 \leq k, l \leq m-1$. Define the polynomials

$$\begin{aligned}\delta(n) &= n(n+1)(n+2)^3(n+3)^5(2n-1)^2(2n+1)^2(2n+3)^2(2n+5)^3\\ &\quad \times (2n+7)^9(4n-1)(4n+1)(4n+3),\\ Q(n,k,l) &= (k+l+2n)(k+l+2n+1)\\ &\quad \times \prod_{x\in\{k,l\}} (x+2n)(x+2n+1)(x+2n+2)(x+3n)(x+3n+2)\prod_{i=0}^{3}(x+4n+i).\end{aligned}$$

There is an explicit polynomial $\widetilde{U}(n,k,l) \in \mathbb{Q}[n,k,l]$ with 8085 terms, of degree 26 in (k,l) and 58 in n, supplied in `telescoping_certificate.json`. Put

$$\mathtt{u}(n,k,l) = \frac{\widetilde{U}(n,k,l)}{\delta(n)Q(n,k,l)}, \qquad \lambda(n) = 5n^2(5n+1)^{\overline{4}}\frac{g(n+2)}{g(n+1)},$$

$$G(n,k,l) = \frac{H(n+2,k,l)\,(n+1+k)\,\mathtt{u}(n,k,l)}{\lambda(n)}.$$

For $n \geq 1$ and $k, l \geq 0$, $\delta(n) \neq 0$ and $Q(n,k,l) > 0$, so G is well defined. $G(n,k,l) = 0$ when $k \geq n+2$ or $l \geq n+2$.

Lemma 10 (telescoping)**.** *For all integers $n \geq 1$ and $k, l \geq 0$,*

$$T(n+1,k,l) - T(n,k,l) = G(n,k+1,l) - G(n,k,l) + G(n,l+1,k) - G(n,l,k). \tag{4.1}$$

Proof. If $k \geq n+2$ or $l \geq n+2$, every term vanishes. Let $0 \leq k, l \leq n+1$ and put $\Pi_1 = (n+1-k)(n+1+k)(n+1-l)(n+1+l)$, $\Pi_0 = (n-k)(n+k)(n-l)(n+l)\Pi_1$. The factorial identities

$$\chi_n(k) = \chi_{n+2}(k)(n-k)(n+k)(n+1-k)(n+1+k), \qquad \chi_{n+1}(k) = \chi_{n+2}(k)(n+1-k)(n+1+k),$$

and $\chi_{n+2}(k+1) = -\chi_{n+2}(k)(n+1-k)/(n+2+k)$ hold throughout this range; at $k = n, n+1$ both sides vanish where appropriate. They show that (4.1) is $H(n+2,k,l)/\lambda(n)$ times

$$\begin{aligned} &-(n+1)^2 \frac{\Pi_0 P(n,k,l)}{D_n(k) D_n(l)(2n+k+l)} + 5n^2(5n+1)^{\overline{4}} \frac{\Pi_1 P(n+1,k,l)}{D_{n+1}(k) D_{n+1}(l)(2n+2+k+l)} \\ &= -(n+1-k)\mathsf{u}(n,k+1,l) - (n+1+k)\mathsf{u}(n,k,l) - (n+1-l)\mathsf{u}(n,l+1,k) - (n+1+l)\mathsf{u}(n,l,k). \end{aligned} \tag{4.2}$$

Indeed, for $j = 0, 1$,

$$T(n+j,k,l) = H(n+2,k,l)\,\frac{g(n+j)}{g(n+2)}\,\Pi_j\,\frac{P(n+j,k,l)}{D_{n+j}(k) D_{n+j}(l)(2n+2j+k+l)},$$

$$\frac{g(n)}{g(n+1)} = \frac{(n+1)^2}{5n^2(5n+1)^{\overline{4}}},$$

and on the right $H(n+2,k+1,l)(n+2+k) = -H(n+2,k,l)(n+1-k)$. Identity (4.2) is an identity of rational functions. After multiplication by $\delta(n)$ and by the product of the 33 distinct linear forms (with multiplicity) that occur in its denominators, it becomes a polynomial identity in $\mathbb{Q}[n,k,l]$. Both sides of that identity have 23800 terms, and they coincide; the program `verify_telescoping.py` checks this exactly. All denominators are positive for $n \geq 1$, $k, l \geq 0$, so the rational identity may be evaluated pointwise. □

Summing (4.1) over $0 \leq k, l \leq n+2$, the right side telescopes; only the terms with first argument 0 survive. With $\Gamma(n) = \sum_{l \geq 0} G(n,0,l)$ we obtain

$$D(n+1) - D(n) = -2\,\Gamma(n) \qquad (n \geq 1). \tag{4.3}$$

Remark 11. The telescoper is the simplest possible: $T(n+1,\cdot,\cdot) - T(n,\cdot,\cdot)$ is a pure two-dimensional difference. The double sum $D(n)$ itself does not satisfy a short recurrence: guessing from 700 terms finds no recurrence of order at most 5 with coefficients of degree below 80 (an empirical observation, not used in the proof), whereas a_n satisfies one of order 3. The boundary terms produced by (4.3) and (3.3) cancel this excess.

5 The single-sum identity

Write $s(n) = a_{n+1} - a_n$ and

$$\beta_n(l) = \frac{(-1)^l}{(n+1-l)!\,(n+1+l)!}, \qquad \nu(n) = \frac{g(n+1)}{(n+1)!^2}.$$

Then $H(n+2,0,l) = \frac{g(n+2)}{(n+1)!^2}\beta_n(l)$. Expressing $T(n,0,l)$, $T(n+1,0,l)$ and $G(n,0,l)$ through $\beta_n(l)$ and using (3.3) and (4.3) gives, for $n \geq 1$,

$$s(n) = \nu(n)\Big[\sum_{l=0}^{n+1} \beta_n(l)\sigma(n,l) + \beta_n(0)\tau(n)\Big], \tag{5.1}$$

where

$$\sigma(n,l) = -\frac{8(n+1)\mathsf{u}(n,0,l)}{5n^2(5n+1)^{\overline{4}}} - 4\frac{(n+1)^2((n+1)^2-l^2)P(n+1,0,l)}{D_{n+1}(0)D_{n+1}(l)(2n+2+l)} + \frac{4(n+1)^4(n^2-l^2)((n+1)^2-l^2)P(n,0,l)}{5(5n+1)^{\overline{4}}D_n(0)D_n(l)(2n+l)}, \tag{5.2}$$

$$\tau(n) = \frac{(n+1)^4P(n+1,0,0)}{2(n+1)D_{n+1}(0)^2} - \frac{(n+1)^6n^2P(n,0,0)}{10n(5n+1)^{\overline{4}}D_n(0)^2}. \tag{5.3}$$

The first term of σ is -8Γ. The second and third are $-4B(n+1)$ and $+4B(n)$; the factors n^2-l^2 and $(n+1)^2-l^2$ make the terms vanish outside the natural ranges. The function τ accounts for $T(n+1,0,0)-T(n,0,0)$. The restriction $\mathsf{u}(n,0,l)$ is a rational function whose numerator has degree 14 in l. Its denominator is a polynomial in n without zeros at integers $n \geq 1$, times linear factors $l+cn+i$ with $c \in \{2,3,4\}$ and $i \geq 0$, all inherited from Q.

Now let $\psi(n) = \nu(n)/U(n)$ with U from (1.2), and $\Phi(n) = s(n)/U(n)$. The ratio

$$r(n) = \frac{\psi(n+1)}{\psi(n)} = -\frac{3(n+1)^2(2n+3)(3n+4)(3n+5)(4n+5)(4n+7)(5n+6)^{\overline{4}}\,q(n)}{(n+2)^2(2n-1)(4n-1)(4n+1)(5n+1)^{\overline{4}}\,q(n+1)}$$

is rational. Put

$$\mathcal{A}(n,l) = r(n)\,\sigma(n+1,l) - (n+2-l)(n+2+l)\,\sigma(n,l).$$

Lemma 12 (Gosper certificate)**.** *Let*

$$\rho(n,l) = \frac{\sum_{e=0}^{30} N_e(n)\,l^e}{\delta_\rho(n)\,R(n,l)}, \qquad R(n,l) = \prod_{i=0}^{3}(l+2n+i)^2 \prod_{i=0}^{5}(l+3n+i) \prod_{i=0}^{7}(l+4n+i),$$

$$\begin{aligned}\delta_\rho(n) = {} & q(n+1)\,n^5(n+1)^3(n+2)^3(n+3)^5(n+4)^5(2n-1)^2(2n+1)^4(2n+3)^3 \\ & \times (2n+5)^3(2n+7)^9(2n+9)^9(3n+2)(4n-1)(4n+1)^2(4n+3)(4n+5)(5n+1)^{\overline{4}},\end{aligned}$$

where the polynomials $N_e \in \mathbb{Q}[n]$ (of degree at most 94) are listed in `gosper_certificate.json`*. The denominator $\delta_\rho(n)R(n,l)$ does not vanish for integers $n \geq 1$, $l \geq 0$. Then*

$$\mathcal{A}(n,l) = -(n+2-l)\,\rho(n,l+1) - (n+2+l)\,\rho(n,l) \tag{5.4}$$

as rational functions, and

$$-\frac{\rho(n,0)}{n+2} + \frac{r(n)\tau(n+1)}{(n+2)^2} - \tau(n) = -\frac{810\,u(n)v(n)(n+1)^7}{(5n+1)^{\overline{5}}}. \tag{5.5}$$

Proof. Both are identities of explicit rational functions. The program `verify_gosper.py` checks them exactly. It recomputes $\mathcal{A}$ from (5.2), clears all denominators in (5.4), and compares the polynomials in $\mathbb{Q}[n,l]$. Identity (5.5) is a univariate rational identity. □

Proposition 13. *For every $n \geq 1$, $\Phi(n+1) - \Phi(n) = V(n)$, and $\Phi(1) = 0$.*

Proof. Since $\beta_n(l) = \beta_{n+1}(l)(n+2-l)(n+2+l)$, (5.1) gives

$$\Phi(n+1) - \Phi(n) = \psi(n)\Big[\sum_{l\geq 0} \beta_{n+1}(l)\mathcal{A}(n,l) + r(n)\beta_{n+1}(0)\tau(n+1) - \beta_n(0)\tau(n)\Big].$$

Next, $\beta_{n+1}(l+1) = -\beta_{n+1}(l)\frac{n+2-l}{n+3+l}$ for $0 \leq l \leq n+2$. Together with (5.4) this gives $\beta_{n+1}(l)\mathcal{A}(n,l) = Z(l+1) - Z(l)$ with $Z(l) = \beta_{n+1}(l)(n+2+l)\rho(n,l)$. Here $\rho(n,l)$ is finite for

$l \geq 0$, and $Z(l) = 0$ for $l \geq n+3$. Hence the sum equals $-Z(0) = -\beta_{n+1}(0)(n+2)\rho(n,0)$. With $\beta_{n+1}(0) = \beta_n(0)/(n+2)^2$, identity (5.5) turns the bracket into $-810u(n)v(n)(n+1)^7\beta_n(0)/(5n+1)^{\overline{5}}$. Finally

$$\psi(n)\beta_n(0) = -\frac{4}{27}(-1)^n \frac{(5n+1)^{\overline{5}}\,(3n)!}{120\,(n+1)^4(n+1)!^3\,u(n)},$$

and since $\frac{4\cdot 810}{27\cdot 120} = 1$ the result is $V(n)$. Lastly, $\Phi(1) = (a_2 - a_1)/U(1) = 0$. □

6 Proof of the main results

Proof of Theorem 1. By Proposition 13 and induction, $\Phi(n) = \sum_{i=1}^{n-1} V(i)$ for $n \geq 1$. Since $U(n) \neq 0$, this gives $a_{n+1} - a_n = U(n)\sum_{i=1}^{n-1} V(i)$. Summing from 1 to $n-1$ and using $a_1 = 1$ gives (1.3). □

Proof of Corollary 2. Write $S(n) = \sum_{i<n} V(i)$, so that $a_{n+1} - a_n = U(n)S(n)$ and $S(n+1) = S(n) + V(n)$. Let $Q_i = p_i + \cdots + p_3$. Since $\sum_j p_j = 0$ identically,

$$\sum_{j=0}^{3} p_j(n)a_{n+j} = S(n)\sum_{i=1}^{3} Q_i U(n+i-1) + Q_2 U(n+1)V(n) + Q_3 U(n+2)\big(V(n)+V(n+1)\big).$$

Divide the first sum by $U(n)$ and the remaining terms by $U(n)V(n)$. The ratios $U(n+1)/U(n)$ and $V(n+1)/V(n)$ are rational, and both quotients vanish identically; `verify_recurrence.py` checks this exactly. The leading coefficient $p_3(n)$ is a product of linear factors positive for $n \geq 1$ and of $p(n) > 0$ (Appendix B). □

7 Computations and checks

How the certificates were found. Both certificates are instances of creative telescoping and Gosper's algorithm [5]; they were computed by linear algebra over prime fields. For the telescoping relation, we chose an ansatz $\widetilde{U}/(\delta Q)$ with symmetric partner $\mathtt{u}(n,l,k)$ and solved the linear conditions at random points (k,l) for each fixed n modulo a prime. A canonical (row-echelon) gauge was used. The values were interpolated in n, combined over 12 primes, and lifted to $\mathbb{Q}$ by rational reconstruction. The order-one telescoper $T(n+1) - T(n)$ was itself discovered this way. The Gosper certificate was found similarly, using 13 primes; the integer lift was stable when one prime was held out. How the certificates were found plays no role in the proof; only the exact checks matter.

Exact checks. The package contains the following programs (Python with SymPy; ≈ 5 minutes in total):

- `verify_telescoping.py`: identity (4.2) in $\mathbb{Q}[n,k,l]$. It also checks (4.1) pointwise in exact rational arithmetic on the full support for $1 \leq n \leq 12$, and (4.3).
- `verify_gosper.py`: identities (5.4) and (5.5), the absence of poles at integers $n \geq 1$, $l \geq 0$, and formula (5.1) against $a_{n+1} - a_n$ for $n \leq 20$.
- `verify_recurrence.py`: Corollary 2 from Theorem 1, and the recurrence and (1.3) against the OEIS table for $n \leq 100$.
- `pfaffian_sum.py`: the fifteen pairing terms and (3.1); the kernel formula (2.2) by symbolic integration for $N \leq 8$; the Pfaffian form of J_N against iterated integration for $N \leq 6$.

- `count_arrays.py`: three independent counts: literal enumeration ($n \leq 2$), dynamic programming over cell bitmasks ($n \leq 7$) and the profile walk ($n \leq 30$). They agree with Theorem 9 and with the OEIS table.

The finite checks detect implementation errors; the proof of the infinite statements consists of the exact identities above.

AI statement. This work was carried out by Claude Opus 5.5, an AI system, including the derivations, the computer-algebra computations and the manuscript. It has been checked by internal review and by the exact computations listed above. It has not been externally peer reviewed and has not been formalised in a proof assistant.

A The fifteen pairing terms

With $N = n-1$ and $\pi = \{(r,6),(p_1,q_1),(p_2,q_2)\}$, the contribution $\operatorname{sgn}\pi \prod_s E_s^{-1}$ of each matching is:

$r;\ (p_1q_1)(p_2q_2)$	$\operatorname{sgn}\pi \prod_s E_s^{-1}$
$5;\ (12)(34)$	$\dfrac{1}{40(N+1)^3(N+k+1)(3N+l+3)}$
$4;\ (12)(35)$	$-\dfrac{1}{10(N+1)^2(N+k+1)(3N+l+3)(4N+l+4)}$
$3;\ (12)(45)$	$\dfrac{1}{30(N+1)^3(N+k+1)(4N+l+4)}$
$5;\ (13)(24)$	$-\dfrac{1}{20(N+1)^2(N+k+1)(3N+l+3)(2N+k+l+2)}$
$4;\ (13)(25)$	$\dfrac{1}{5(N+1)(N+k+1)(3N+l+3)(4N+l+4)(2N+k+l+2)}$
$2;\ (13)(45)$	$-\dfrac{1}{15(N+1)^2(N+k+1)(2N+k+2)(4N+l+4)}$
$5;\ (14)(23)$	$\dfrac{1}{20(N+1)^2(N+k+1)(3N+k+3)(2N+k+l+2)}$
$3;\ (14)(25)$	$-\dfrac{1}{5(N+1)(N+k+1)(4N+l+4)(2N+k+l+2)(3N+k+l+3)}$
$2;\ (14)(35)$	$\dfrac{1}{5(N+1)(N+k+1)(2N+k+2)(4N+l+4)(3N+k+l+3)}$
$4;\ (15)(23)$	$-\dfrac{1}{5(N+1)(N+k+1)(3N+k+3)(4N+k+4)(2N+k+l+2)}$
$3;\ (15)(24)$	$\dfrac{1}{5(N+1)(N+k+1)(4N+k+4)(2N+k+l+2)(3N+k+l+3)}$
$2;\ (15)(34)$	$-\dfrac{1}{5(N+1)(N+k+1)(2N+k+2)(4N+k+4)(3N+k+l+3)}$
$1;\ (23)(45)$	$\dfrac{1}{15(N+1)^3(2N+k+2)(4N+l+4)}$
$1;\ (24)(35)$	$-\dfrac{1}{5(N+1)^2(2N+k+2)(4N+l+4)(3N+k+l+3)}$
$1;\ (25)(34)$	$\dfrac{1}{5(N+1)^2(2N+k+2)(4N+k+4)(3N+k+l+3)}$

Their sum, symmetrised in $k \leftrightarrow l$, equals (3.1). The factor $3N+k+l+3$ cancels in the symmetrisation.

B The recurrence

The recurrence of Corollary 2 is $\sum_{j=0}^{3} p_j(n) a_{n+j} = 0$ with

$$p_3(n) = 3(n+2)^2(n+3)^4(2n+3)(2n+5)^2(3n+7)(3n+8)(4n+9)(4n+11)\,p(n),$$
$$p_0(n) = 25(2n-1)^2(2n+1)(4n-1)(4n+1)(5n+1)^{\overline{4}}(5n+6)^{\overline{4}}\,p(n+1),$$

where $p(n+1)$ expands to the undecic printed in [4], and p_1, p_2 are the degree-24 polynomials of the OEIS entry (both stored in expanded form in `data/oeis_recurrence.json`). Exactly: $p_2(n) = -(n+2)^2(2n+3)\,\pi_{21}(n)$ and $p_1(n) = -15(2n+1)(5n+6)^{\overline{4}}\,\pi_{19}(n)$, where π_{21} and π_{19} are the polynomials of degree 21 and 19 given there. The proof uses only $\sum_j p_j = 0$ and the two vanishing quotients of Section 6, which are checked for these exact polynomials.

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